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| **Fiscal Year:** | FY 2004 | **Task Last Updated:** | FY 03/24/2006 |
| **PI Name:** | Griffin, DeVon Ph.D. | | |
| **Project Title:** | Phase-Shifting Point Diffraction Interferometer for Microgravity Fluid Physics | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | FLUID PHYSICS | | |
| **Program/Discipline--Element/Subdiscipline:** | FLUID PHYSICS--Complex fluids | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **City:** | Cleveland | **State:** | OH |
| **Zip Code:** | 44135 | **Congressional District:** | 10 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 98-HEDS-03 |
| **Start Date:** | 03/01/2000 | **End Date:** | 10/30/2004 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 5 | **Monitoring Center:** | NASA GRC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Marshall, Kenneth ( Lanoratory for Laser Energetics )<br>Mercer, Carolyn ( NASA Glenn Research Center ) | | |
| **Grant/Contract No.:** | IH10113087 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Previous phase-shifting LCPDIs have not operated near video frame rates due to difficulties in switching the optical path in the liquid crystal cell. Inaccuracies were also present in the device due to hysteresis in optical path upon switching and electrical-field dependence of the dyes used in the device. We believe that we can overcome most of these problems through a judicious selection of dye in the cell. We will do this through a search and characterization of available commercial dyes and possible synthesis of our own. The objective of this work is to develop a phase-shifting LCPDI that operates at video frame rates. Such an interferometer is compact enough to be held in the hand and extraordinarily resistant to vibration. Most interferometers require a large volume to operate and are quite sensitive to environmental vibration. Both of these problems will be eliminated if we are successful. As a result, we will deliver a device that will facilitate fluid physics and other research in the International Space Station as well as provide a tool for ground-based investigations that require interferometry, regardless of discipline. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | |
| **Task Progress:** | No progress report this period. |
| **Bibliography Type:** | Description: (Last Updated: 02/27/2004) |